

AI MOBILE APP FOR EARLY IDENTIFICATION OF STUDENT MENTAL HEALTH VIA BEHAVIOR & SPEECH

Project ID: 25-26J-278

Project Proposal Report

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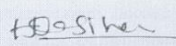
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DECLARATION

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

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Date: 29.08.2025

Signature of the co-supervisor:

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Ms. Ishara Weerathunga

Date:

ABSTRACT

As mobile devices and the use of smart technologies become a larger percentage of each day, people produce increasingly high amounts of multivocal data in the form of speech, written words, and behavioral interactions. Although such data streams are valuable, they can cause information overload as they are unstructured and most of the time the user can not pick meaningful patterns since they are in abundance. This project is an on-device AI summarizing framework that securely gathers, preprocesses and fuses speech transcripts, journal text, and behavioral logs on-device. The system is designed in four significant units Data is obtained via mobile microphones, user-entered text and system application program interfaces with user consent. Second, preprocessing cleans the data by removing noise in audio, normalizing text inputs, and summarizing behavioral logs into summaries that are relevant to query the data. Third, a trend-detecting lightweight large language model (LLM), tuned to mobile hardware by quantization and distillation, incorporates multimodal data to identify trends on emotional tone, stress, and productivity changes. Lastly, summarization and storage create concise, daily summaries, and encrypt them safely and store them locally disposing of raw sensitive information. This application will serve as an early detection application and not a diagnostic tool. Notifying users when there is concern and providing self-care content and suggestions to see a mental health professional is important.

Key Words – Quantization, Distillation, Multimodal data, Trend-detecting, Concise

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LIST OF ABBREVIATIONS

Abbreviation	Description
ML	Machine Learning
AI	Artificial Intelligence
LLM	Large Language Model
ASR	Automatic Speech Recognition
DL	Deep Learning
UI	User Interface
UX	User Experience
STT	Speech-to-Text
BERT	Bidirectional Encoder Representations from Transformers

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1. INTRODUCTION

In the past few years, stress, depression, and anxiety have developed into some of the most significant global public health issues. The pressures of contemporary life, exposure to digital devices, and new social and work environments are contributing to the prevalence of these disorders in people of all ages. If these disorders are left unchecked, they may result in reduced productivity, cognitive impairment, damage to interpersonal relationships, and the onset of more severe mental and physical health issues. Therefore, identification and intervention at an early stage are crucial in reducing the impact these disorders have on an individual. Unfortunately, mental health detection in most cases still relies on self-evaluation forms, clinical examinations, or behavior observation. These approaches, though helpful, are often hindered by stigma, lack of accessible professionals, symptom underreporting, episodic nature of clinical evaluation, and many others. Because of this, chances to accurately and promptly recognize mental health problems are so often lost.

Artificial Intelligence (AI) offers the most significant opportunity to address these gaps and to enable more objective, continuous, and data-driven evaluation of mental health and psychological well-being. AI systems equipped with advanced algorithms in machine learning and natural language processing can analyze large amounts of multimodal data to find subtle markers of stress, depression, and anxiety. For example, variations in voice tone, speech rhythm, and language usage can reflect emotional states; physiological signals such as heart rate variability or sleep patterns captured by wearables can indicate stress levels; and text-based communications may contain linguistic patterns correlating with depressive or anxious thought processes. These signals, when integrated and analyzed in real time, allow AI models to build comprehensive behavioral and emotional profiles that support more accurate detection than single-source indicators alone.

Direct deployment of optimized models on consumer devices is now possible thanks to developments in edge AI and mobile processors. On-device summarization analyzes written text, behavioral logs, and speech transcripts without sending sensitive

information outside the device by using lightweight large language models (LLMs) like DistilBERT or quantized transformers [7,8].

To generate succinct summaries, this study suggests an on-device AI summarization framework that incorporates multimodal data. The objective is to guarantee data privacy and efficiency while offering safe, real-time, and context-aware insights into users' everyday lives. Applications for such a system are numerous and include productivity improvement, education, mental health, and digital well-being.

1.1 Background & Literature Survey

An increasing amount of research shows how speech, text, and behavioral data can be used to better understand digital well-being and mental health. In their study of the relationships between mobile sensor data and depressive symptoms, Saeb et al. [1] demonstrated that everyday behaviors like location variability, sleep patterns, and physical activity levels can serve as useful stand-ins for psychological states. Building on this, Mohr et al. [3] introduced the concept of personal sensing, emphasizing how ubiquitous smartphone sensors and machine learning can provide continuous, real-world monitoring of mental health. All of these studies demonstrate how digital traces incorporated into routine behaviors provide important information about anxiety, depression, and stress.

Linking linguistic and acoustic characteristics to psychological states has advanced significantly in the speech domain. According to research reviewed by Cummins et al. [2], there are significant correlations between the degree of depression or the risk of suicide and prosodic markers like pitch, tone, fluency, and speech pauses. By using deep convolutional recurrent neural networks for end-to-end emotion recognition, Trigeorgis et al. [6] went one step further in this regard. Their research showed that discriminative features can be automatically learned from unprocessed speech signals by contemporary neural architectures, eliminating the need for manually created features and greatly increasing prediction accuracy. These contributions demonstrate that speech data is a potent modality for mental health monitoring and can be easily recorded by wearables and smartphones.

Signals based on text and interaction have also shown promise in identifying depressive and stress-related behaviors. Depressive states in written and spoken communication have been associated with linguistic markers, including negative sentiment, decreased lexical diversity, and increased use of first-person pronouns. Additional insights can be gained from behavioral features in digital interactions in addition to language content. For example, Ghosh et al. [4] examined keystroke dynamics and discovered that error frequency, pauses, and typing speed could all be markers of stress and mood swings. These results highlight the diversity of implicit (behavioral) and explicit (linguistic) data streams in digital settings.

1.2 Research Gap

Although earlier studies have demonstrated the benefits of examining speech, text, and behavioral data to gain understanding of productivity and mental health, there are still several important gaps that need to be filled.

First, a single-modality focus has been adopted by a large portion of the literature currently in publication. For example, a lot of research has been done on speech characteristics to detect depression [2], on typing and keystroke dynamics to identify stress [4], and on behavioral analysis using mobile app usage or sensor data [1], [5]. These pieces show the potential of individual modalities, but they don't take advantage of multimodal integration's complementary qualities. A more comprehensive picture of a person's psychological state could be captured by combining speech, text, and behavioral cues, eliminating the noise and ambiguity that come with depending solely on any single signal source.

Second, for computationally demanding model inference, most existing implementations mainly depend on cloud-based infrastructures [3], [10]. Cloud processing presents significant issues with latency, reliance on constant internet connectivity, and above all data privacy, despite providing scalability and access to large AI models. Private conversations, journal entries, and behavioral logs are

examples of sensitive data that must be sent to distant servers, raising the possibility of abuse, security lapses, or illegal access. This erodes user confidence and prevents such systems from being widely used in real-world situations.

Third, research on lightweight summarization frameworks that integrate multimodal inputs directly on-device is still lacking, despite recent developments in transformer-based architectures [7], [8] and the rise of mobile-optimized AI frameworks like TensorFlow Lite and Core ML. Current mobile AI apps either rely on cloud-based summarization pipelines or concentrate on specific recognition tasks (like activity recognition [9]). As a result, there is still much untapped potential for creating effective, privacy-preserving, real-time systems that combine speech, text, and behavior into succinct insights.

The lack of a multimodal, on-device AI summarization framework that considers computational limitations while guaranteeing security, personalization, and the production of actionable insights is therefore the main research gap.

1.3 Research Problem

How can speech, text, and behavioral data be directly summarized on mobile devices using a lightweight, multimodal AI framework that offers safe, instantaneous, and context-aware insights without sacrificing device performance or privacy?

In order to advance edge AI applications and create systems that protect user privacy while providing actionable, multimodal insights in fields like education, mental health, and digital well-being, this issue must be resolved.

2.OBJECTIVES

2.1 Main Objective

The main goal of this study is to develop and assess a multimodal, on-device, privacy-preserving AI summarization framework that combines behavioral, textual, and speech data to deliver succinct, real-time insights into anxiety, depression, and stress. The suggested system will function locally on wearable and mobile devices, in contrast to traditional cloud-based solutions, protecting user privacy while resolving issues with latency, bandwidth usage, and offline usability. The goal of the research is to help create useful, safe, and effective digital mental health solutions by utilizing recent developments in lightweight AI models and mobile deployment frameworks.

2.2 Specific Objectives

1. To evaluate the function of behavioral, textual, and speech data in identifying stress, anxiety, and depression while determining each condition's unique advantages and disadvantages.
2. To create a multimodal fusion strategy that combines these modalities to provide a more thorough and precise summary of mental health.
3. To create AI models that are lightweight and appropriate for on-device deployment while considering the memory and processing limitations of wearable and mobile devices.
4. To put in place privacy-preserving measures that guarantee private information is handled safely and stays on the device.
5. To compare the suggested system with traditional cloud-based methods in order to assess its accuracy, efficiency, latency, and user trust.
6. To investigate the framework's practical uses in the fields of education, workplace efficiency, health monitoring, and individual digital wellbeing.

3. METHODOLOGY

The goal of the suggested research methodology is to create, put into practice, and assess a multimodal, privacy-preserving, on-device AI summarization framework for identifying anxiety, depression, and stress. Data collection and preprocessing, feature extraction, multimodal fusion, on-device deployment, and evaluation are the five main phases of the methodology. Every step has been meticulously planned to guarantee that the system is precise, computationally effective, and compliant with user privacy ethics.

1. Gathering and Preparing Data

A variety of data streams, including speech, text, and behavioral signals, will be gathered in order to create strong multimodal models:

Voice notes, meetings, and conversational audio recorded by smartphones and wearable microphones will all be considered speech data. Applying noise reduction techniques, guaranteeing a wide range of acoustic conditions, and collecting data in a naturalistic manner are important factors. User-generated digital communication, such as brief messages, journal entries, or spoken content transcriptions, will serve as the source of text data. Tokenization, stop-word elimination, and normalization are examples of preprocessing that will get inputs ready for models used in natural language processing.

Mobile and wearable sensors provide behavioral data, identifying trends like keystroke dynamics, app usage, mobility, and sleep/activity cycles. To lessen user-specific biases, data will be normalized and anonymized. Participants will give their informed consent, and all sensitive data will be processed locally and encrypted to maintain ethical compliance. To further lower the risk of re-identification, different privacy strategies could be used.

2. Feature Extraction

Specialized feature extraction techniques will be used for every modality:

Speech characteristics: Mel-Frequency Cepstral Coefficients (MFCCs), pitch, jitter, and prosody are examples of traditional acoustic characteristics that will be extracted. Additionally, transformer-based speech encoders that have already been trained will be used to obtain deep representations.

Text features: Cutting-edge models for natural language processing, like BERT, DistilBERT, or comparable lightweight transformer models, will be used. DistilBERT is especially well-suited for on-device deployment because of its efficiency and accuracy balance. Additionally, linguistic indicators like sentiment polarity, semantic coherence, and pronoun usage will be included.

Behavioral features: Statistical and temporal descriptors will be used to represent keystroke timing, app usage statistics, mobility metrics, and circadian rhythm patterns. Subtle behavioral changes linked to stress, depression, and anxiety are captured by these characteristics.

To enable multimodal integration, all features will be standardized and positioned within a shared temporal window.

3. Summarization and Multimodal Fusion

Integrating diverse modalities into a single summarization framework forms the basis of the methodology. We'll look at two fusion tactics:

Early Fusion: All modalities' raw or low-level features are combined and fed into a single deep learning model. Although it may require a lot of computing power, this method captures cross-modal correlations.

Late Fusion: Each modality's predictions are combined using ensemble techniques (e.g., weighted voting or attention mechanisms) after independent models have been trained for it. This method enables modular design while reducing complexity.

We will also examine a hybrid strategy that feeds modality-specific encoders into an attention-based fusion network, combining the advantages of both approaches. Short insights like "High stress detected during morning communication" or "Consistent reduction in activity linked to depressive tendencies" will then be produced by the summarization component.

4. Device Deployment

Modifying the multimodal summarization system for on-device execution is a significant methodological contribution. The following optimizations will be used in order to address memory and computational limitations:

Lightweight Frameworks: To implement deep learning models on smartphones and wearables, deployment will make use of mobile AI toolkits like PyTorch Mobile, Core ML, or TensorFlow Lite.

Edge Optimization: To further cut down on energy usage, cache intermediate representations and use event-driven processing (i.e., only processing when new speech, text, or behavioral data is detected).

To ensure that speech, text, and behavioral logs stay on the user's device, all sensitive processing will take place locally. To improve data protection, secure enclaves and on-device encryption will be taken into consideration.

5. Evaluation Framework

The suggested system will be assessed from several angles:

Accuracy and Effectiveness:

The accuracy of stress, depression, and anxiety detection will be assessed using performance metrics like precision, recall, F1-score, and Area Under the Curve (AUC). In addition to expert review, established metrics like ROUGE and BERTScore will be used to evaluate the quality of the summaries.

Efficiency and Scalability:

In order to measure latency, memory usage, and energy consumption, computational performance will be evaluated under various device constraints. The system will be tested with different input lengths and data loads in order to assess scalability.

Security and Privacy:

Threat modeling and data leak scenario simulation will be used to verify privacy-preserving features. Surveys and usability studies will be used to assess user trust and perceived safety.

User-Centered Assessment:

Pilot studies will be carried out in real-world settings, including health monitoring, workplace productivity, and education. The system's usability, the interpretability of the summaries, and the perceived advantages for digital well-being will all be evaluated.

Ethical Considerations.

Ethical compliance is essential to the methodology because mental health data is sensitive. Prior to analysis, anonymization will be implemented, and each participant will give their informed consent. With opt-in and opt-out options, users will have complete control over their data. In order to avoid making unfair or discriminatory predictions, bias in AI models will be continuously monitored.

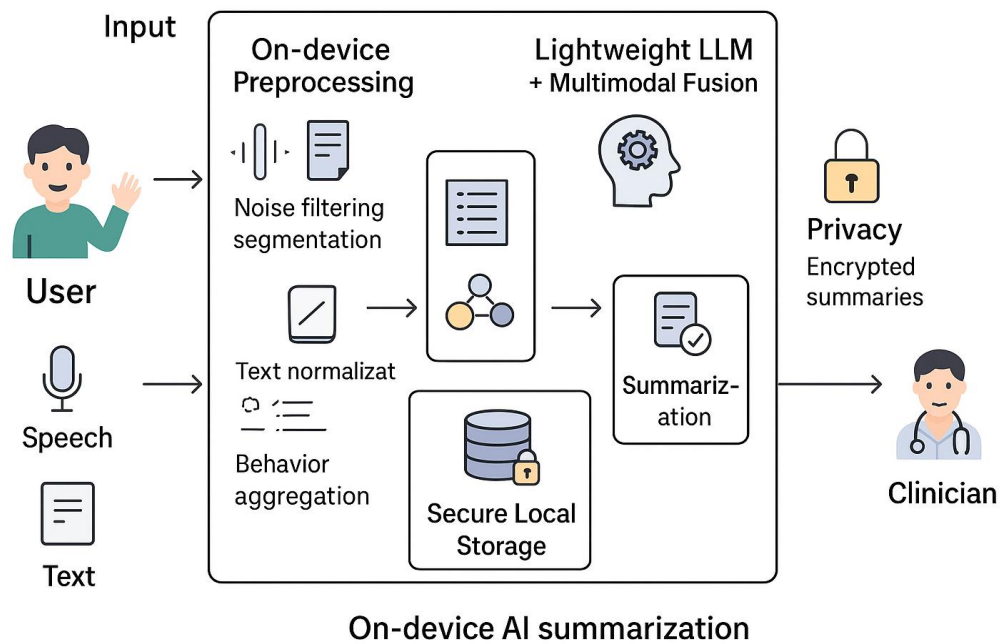


Figure 1: High Level Architecture

4.SYSTEM REQUIREMENTS

A number of functional, non-functional, and ethical requirements must be met for the development and implementation of a multimodal, on-device AI summarization framework for the detection of stress, depression, and anxiety to be successful. These guarantee that the framework is efficient, technically possible, and compliant with user trust and privacy standards.

1. Functional Requirements

Multimodal Data Collection: Speech, text, and behavioral data streams from wearable sensors and smartphones must be captured by the system.

Preprocessing Capabilities: To guarantee compatibility across modalities, raw data needs to be cleaned, normalized, and temporally aligned.

Feature Extraction: In order to detect mental health issues, the system needs to extract behavioral, linguistic, and acoustic characteristics.

Summarization Module: AI models must produce succinct, comprehensible summaries that highlight important trends and possible markers of mental health.

Processing on-device: All summarization and inference must take place locally, independent of external servers.

2. Non-Functional Requirements

Efficiency: Models need to be tuned for mobile devices' small memory footprint, low latency, and low battery usage.

Scalability: There should be minimal performance degradation when the framework manages different data types and input lengths.

Robustness: The system must work with a variety of user demographics and in noisy

environments (such as background noise or erratic typing patterns).

Interoperability: Frameworks such as TensorFlow Lite or Core ML must be used to support compatibility with various mobile platforms, including wearable OS, iOS, and Android.

Reliability: Throughout prolonged use, the system must reliably produce accurate insights.

3. Ethical and Security Requirements

Preserving privacy: All private text, voice, and behavior logs be encrypted and processed locally.

User Autonomy: With opt-in/opt-out choices, users must maintain control over data collection.

Bias Mitigation: To prevent biased results across demographic groups, the system's fairness must be assessed.

Transparency: To build user trust, summaries need to be comprehensible and interpretable.

5.BUDGET AND BUDGET JUSTIFICATION

Table 5.1 provides an overall budget for the overall proposed system.

Expenses	Requirement Cost (Rs.)	Justification
Travelling cost for surveys and awareness programs	15,000.00	To conduct initial data collection, distribute questionnaires, and raise awareness about burnout among students at universities.
Data Collection (standardized questionnaires, survey tools, digital forms)	10,000.00	Required for acquiring reliable datasets from students for training, testing, and validating the system.
Deployment cost (server hosting, backend services)	10,796.00 / month	To maintain system availability and reliability through cloud-based deployment.
Hosting cost in Play Store	8,076.00	One-time cost to publish the mobile application on the Google Play Store for student access.
Miscellaneous (printing, promotional activities, student awareness sessions)	5,000.00	For marketing, leaflets, and awareness campaigns to ensure successful adoption of the app.
Total Expenses	48,872.00 + 10,796.00/month	-

6.COMMERCIALIZATION

Target Audience and Market Space

TARGET AUDIENCE

- **Individual Users:** Students, professionals and general smartphone users seeking self-monitoring and well-being insights.
- **Workplaces:** Organizations aiming to support employee mental health, reduce burnout and enhance productivity.
- **HealthCare Providers:** Psychologists, psychiatrists and therapists who can use the app as a supplementary tool for remote monitoring.
- **Educational Institutions:** Universities and schools looking to provide digital well-being resources for students.

MARKET SPACE

- Meditation & Mindfulness Apps (e.g. Headspace, Calm)
- AI Chatbot Therapists (e.g. Woebot, Wysa)
- Cognitive Behavioral Therapy tools (e.g. Talkspace, BetterHelp)

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APPENDICES

Appendix 1: Work Break Down Chart

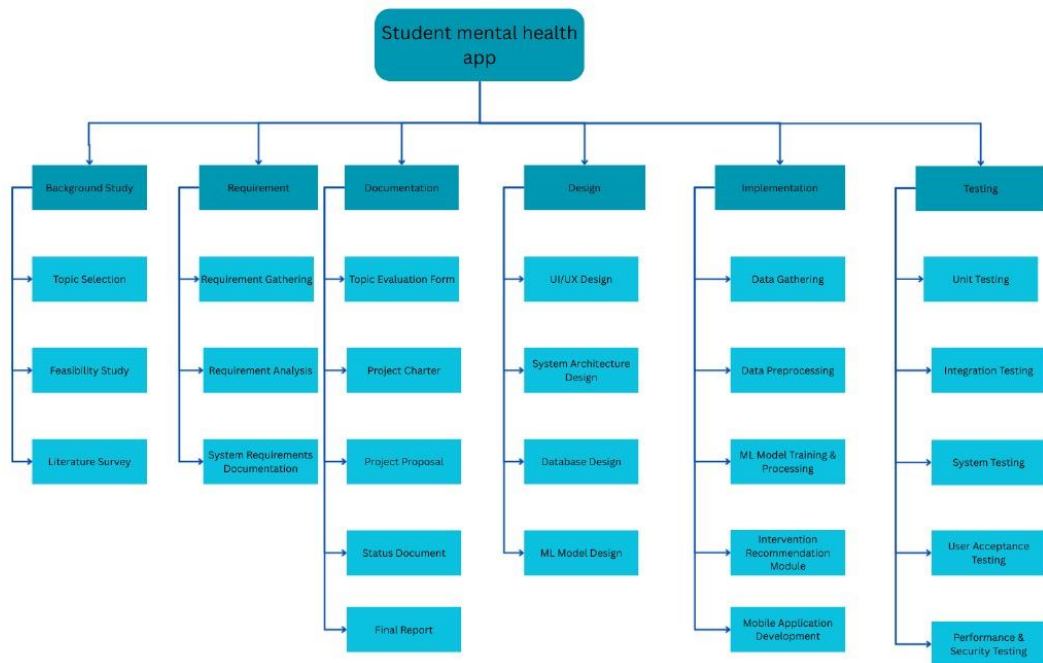


Figure 2: Work Break Down Chart

Appendix 2: Gantt Chart

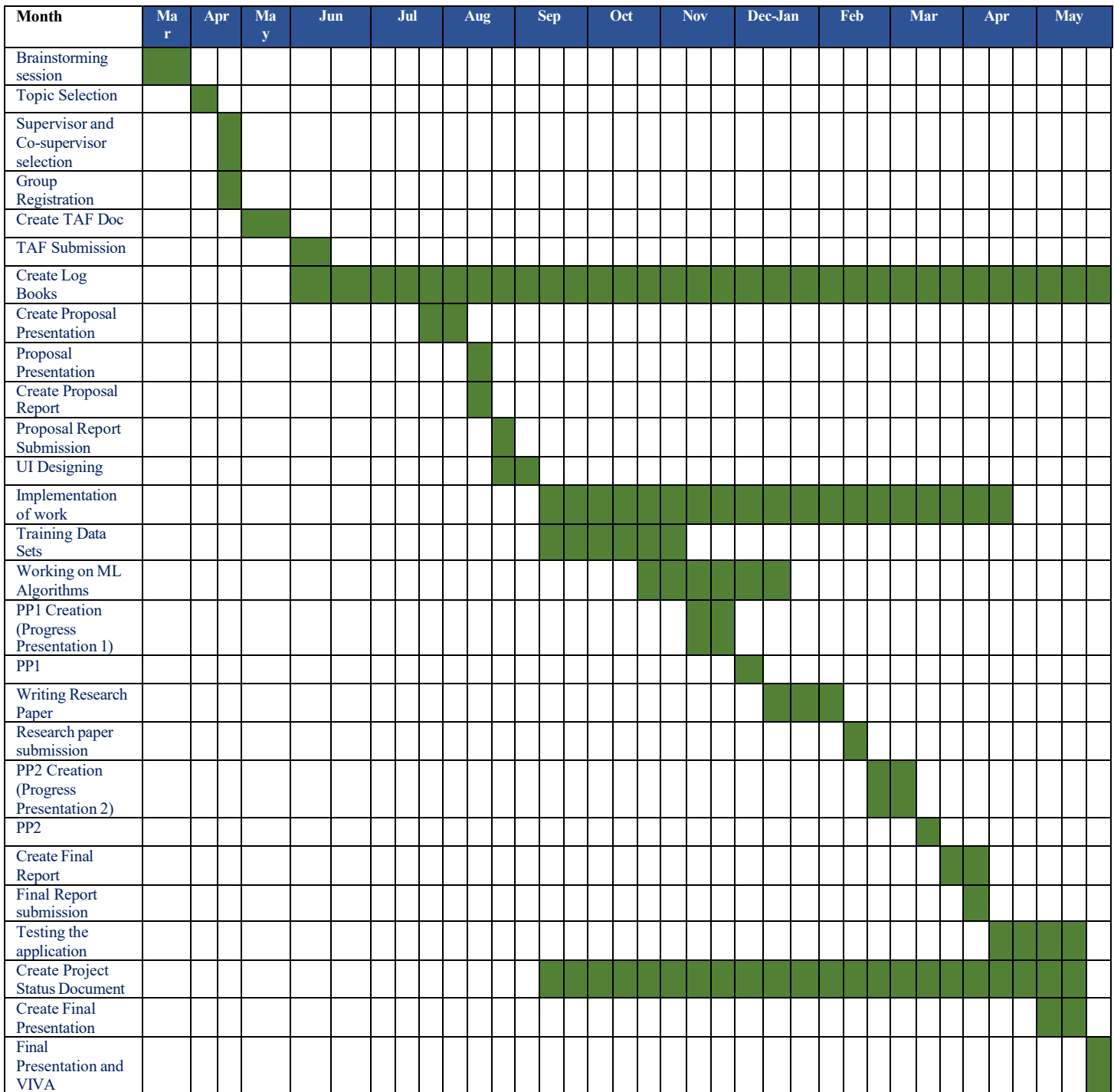


Figure 3: Gantt Chart